

Physics Questions

persons working without the aid of power.

106. An incandescent bulb rated as 100 W at 110 V is connected to a 220 V power supply. The power that dissipates in the bulb would be:

107. If the length of a current carrying wire is halved, for a given potential difference, the current

1. India will reach its non -fossil energy capacity of 500 GW by 2030.

2. India will meet 50% of its energy requirements from renewable energy by 2030.

appointed in 1927 to investigate and clarify the relationship between the paramount power

- He also introduced a regular force of police in 1791 .

powers vested in him, functions mainly of a managerial nature; or

The LCM of 24 and 30 is the product of the highest powers of all the prime factors involved,

The GCD of 36 and 100 is the product of the lowest powers of all the common prime factors,

The units digit of powers of 2 follows a repeating cycle of four: {2, 4, 8, 6}. To find the units digit of 22^{100} ,

Here is a table that shows the units digit of powers of 2:

Power Units digit

As we can see, the units digit of powers of 2 repeats in a cycle of four: {2, 4, 8, 6}. The remainder of 100

Here is a proof of why the units digit of powers of 2 repeats in a cycle of four:

As you can see, the units digit of powers of 2 repeats in a cycle of four: {2, 4, 8, 6}. This is because the units

Atal Vayo Abhyudaya Yojana is a scheme of Ministry of Social Justice and Empowerment.

organization's energy use.

operations can take place concurrently or in parallel.

under any law relating to co -operative societies for the time being in force in any State".

being in force, the Board may be kept under suspension or superseded for a maximum period of six

not given the complete power to company hence called as regulating act.

National Social Security Board to exercise the powers conferred on, and to perform the functions

under the Co-operative Societies Act, 1912, or under any other law for the time being in force in the aid of power .

negative power of 2 and sum them up.

based on their relative position to the current cell. When a formula or function with relative remain fixed or absolute regardless of their position relative to the current cell. Absolute references

When the incandescent bulb rated at 100 W and designed for 110 V is connected to a 220 V power supply, the power that dissipates in the bulb can be determined using the formula:

$$\text{Power (P)} = (\text{Voltage (V)})^2 / \text{Resistance (R)}$$

The power dissipated when bulb is connected to 220 V supply is given by

$$\text{Power} = V^2/R$$

$$\text{Power} = 220^2/121$$

$$\text{Power} = 48400/121$$

$$\text{Power} = 400 \text{ W}$$

Therefore, the power that dissipates in the bulb when connected to a 220 V power supply is:

According to Ohm's Law, the current (I) flowing through a conductor is directly proportional to the current is inversely proportional, when resistance is halved, current doubles .

2 The energy change is usually small. The energy change is usually large.

- First - India will reach its non-fossil energy capacity to 500 GW by 2030.
- Second - India will meet 50 % of its energy requirements from renewable energy by 2030.

like No poverty, Zero hunger, Gender equality, Affordable and clean energy, Industry, innovation,

Chemistry Questions

selecting your answers based solely on the content of the passage and opinion of the author only.

base of side 2 m. If a full tank of water is drained out completely in a long pipe(a) 2 of circular

2. Research and Development based activity comes under quaternary activity .

(a) Global Anti -Base Erosion Rules

(b) A database of the financial performance of companies that is specially designed for academia A

(d) A database in which collection and storing of data is done on user's system and this is basically

1. DNA refers to Deoxyribonucleic Acid.

3. It is composed of Ribonucleic Acid.

(b) All compounds containing hydrogen are acidic.

(c) Separation of H ions from HCl molecules cannot occur in the absence of water.

(d) Water soluble bases are known as alkalis.

(a) blue colour salt

(b) white colour salt

(c) green colour salt

(d) brown colour salt

109. Tartaric acid is found in:

110. Which one of the following is an example of chemical change?

Based on the divisibility by 8, options C and D satisfy this condition.

Based on the divisibility by 11, option D satisfies this condition.

Side of the square base (a) = 2 m

Volume of the tank = Base area $\times$ Height

Base area = side² (since it's a square base)

- This has been done to counter Base Erosion and Profit Sharing Strategies of Multi National Entities (MNEs) and are thus part of the Global Anti -Base Erosion (GloBE) Rules . Hence

- The Global Anti -Base Erosion Rules (GloBE Rules) are a set of rules designed to ensure that

- Base erosion and profit shifting (BEPS) refers to tax planning strategies used by multinational

Prowess is a database of the financial performance of companies. Annual Reports of companies, Prowess database that is specially designed for academia. Pro wessdx facilitates easy downloading

DNA stands for Deoxyribonucleic Acid . It is a molecule that contains the genetic instructions for chemical bonds. The nucleotides are arranged in a specific order, which determines the genetic

- The DNA molecule is shaped like a double helix, which means that it is two strands of

- The nucleotides in DNA are made up of three different molecules: deoxyribose, a sugar;

phosphate; and one of four nitrogenous bases: adenine (A), guanine (G), cytosine (C), or contained in the molecule.

1. Statement 1 is correct: DNA is a long, double -stranded molecule that carries the genetic information in most organisms. It is commonly known as Deoxyribonucleic Acid. Hence

3. Statement 3 is incorrect: DNA is composed of deoxyribonucleic acid , not ribonucleic acid.

DNA and RNA are both nucleic acids but have different sugar components in their structure.

two identical copies of the DNA molecule.

because they are too reactive. They will always combine with other ions or molecules to form acids.

hydrogen are acidic. For example, water, which contains hydrogen, is not acidic. It is actually molecules cannot occur in the absence of water because HCl is a covalent compound, which means that the hydrogen and chlorine atoms are held together by a shared pair of electrons.

hydrogen ions will not be able to separate from the HCl molecules.

- Statement (d) is correct and thus not the correct answer: Water soluble bases are known as alkalis. This is because they have a pH of 7 or more. pH is a measure of how acidic or basic a substance is. A pH of 7 is neutral, a pH of less than 7 is acidic, and a pH of more than 7 is basic. Alkali metals are the six chemical elements that make up Group 1 (Ia) of the periodic table — forms alkalies (i.e., strong bases capable of neutralizing acids).

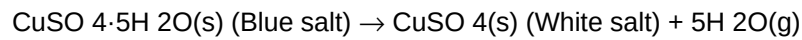
atoms to move more

- The atoms in a metal are held together by the attraction of the nuclei to electrons which are delocalized over the whole metal mass. As the atoms increase in size, the distance between received royal assent in August 1935. This Act was based on Simon Commission Report and references is copied or filled to other cells , the references are adjusted automatically based on adjust based on the relative position. Therefore, it's an incorrect answer.

This is because the atoms in the wire vibrate more as the temperature increases, which

as dehydration. During this process, the water molecules that are bound to the copper sulphate

The chemical equation for the dehydration of copper sulphate pentahydrate can be represented as



are bright blue in colour due to the presence of water molecules. However, as the water molecules

Tartaric acid is a naturally occurring organic acid that is commonly found in tamarind. Tartaric acid

is also found in grapes and wine, where it plays a role in flavor and acidity.

A chemical change is a change in which the chemical composition of a substance is altered. This

changes, the chemical composition of the substances does not change.

magnesium oxide. This is a chemical change. The following reaction occurs when magnesium burns

The differences between chemical and physical changes are given below:

S.No. Physical change Chemical change

Biology Questions

(b) The Plantations Labour Act, 1951

71. Consider the following statements about DNA:

(a) Tap root

(b) Epiphytic root

(c) Fibrous root

(d) Respiratory root

D. Subsistence allowance 4. Employees Provident Fund and Miscellaneous Provisions

Appellate Authority under the Employees' Provident Fund and Miscellaneous Provisions Act,

Miscellaneous Provisions Act, 1952, is not correct?

104. In a worksheet, which one of the following is not a cell referencing style?

living organisms. DNA is made up of two long strands of nucleotides, which are linked together by

information contained in the DNA.

thymine (T). The order of the nucleotides in DNA determines the genetic information

2. Statement 2 is incorrect: DNA is not located in the ribosomes. Ribosomes are cellular

structures involved in protein synthesis. DNA is found primarily in the cell nucleus (nuclear DNA) and, in some cases, in other cellular compartments like mitochondria and chloroplasts . DNA contains deoxyribose sugar, while RNA contains ribose sugar.

4. Statement 4 is correct: DNA has the ability to replicate or make a copy of itself through a process called DNA replication . During replication, the DNA strands separate, and each

Velamen is a spongy tissue that is found on the aerial roots of epiphytic plants . It helps to absorb moisture and nutrients from the air. Epiphytic plants are plants that grow on other plants , but they do not get their nutrients from the host plant. They get their nutrients from the air and from the

In contrast, tap roots, fibrous roots, and respiratory roots do not typically form velamen tissue. Tap roots are the main central root of a plant, fibrous roots are a network of thin roots, and respiratory roots are specialized roots found in certain plant s that grow in waterlogged or anaerobic soil

Section 16 of the Employees' Provident Fund and Miscellaneous Provisions Act, 1952 - Act not to

In a worksheet, the term "real -time referencing" is not a recognized cell referencing style. It does not correspond to any commonly used method of referring to cells in a worksheet.

Relative referencing : Relative referencing is a cell referencing style in which the cell references are

Absolute referencing : Absolute referencing is a cell referencing style in which the cell references

Mixed referencing : Mixed referencing is a cell referencing style that combines relative and absolute is relative. It allows you to anchor a specific part of a cell reference while allowing the other part to

Therefore, the option that is not a recognized cell referencing style is Real -time referencing.